99

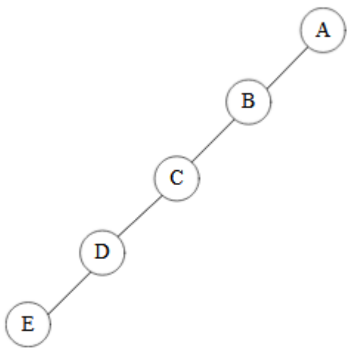
1. (10%) What is the maximum number of nodes in a *k*-ary tree of height *h*? Prove your answer.

prov: $k=2$ 且 $h=3$ 時 ⇒ 共有 $2^0+2^1+2^2$ 個 node
$k=3$ 且 $h=3$ 時 ⇒ 共有 $3^0+3^1+3^2$ 個 node
⇒可推得 k-ary tree of height h 的 node 總數為

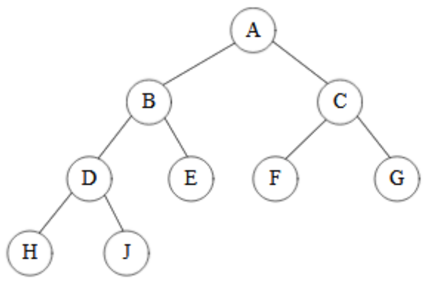
$$k^0+k^1+k^2+\cdots+k^{h-1} = \frac{k^h-1}{(k-1)}$$

Ans: $\frac{k^h-1}{k-1}$ 個

2. (20%) Write out the inorder, preorder, postorder, and level-order travels for the binary trees (a) and (b).



(a)



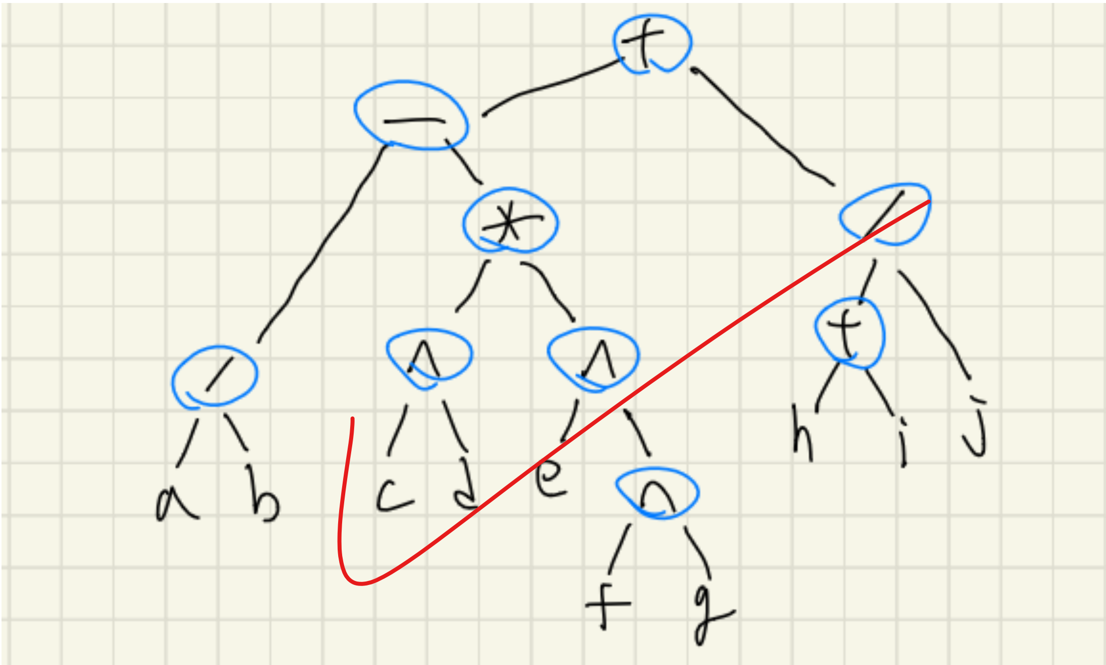
(b)

(a) inorder: EDCBA
preorder: ABCDE
postorder: EDCBA
level order: ABCDE

(b) inorder: HDJBEAFCG
preorder: ABDHJECFG
postorder: HJDEBFGCA
level order: ABCDEFGHJ

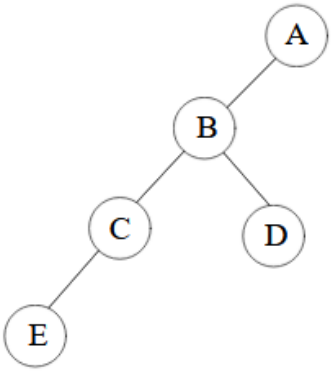
3. (15%) Considering operator precedence, draw the expression tree of the infix expression. Note that ^ is power.

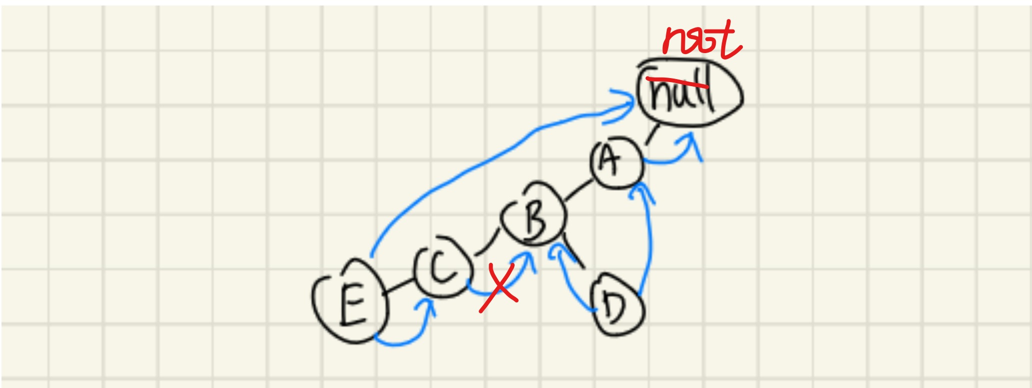
a / b – c ^ d * e ^ f ^g + (h + i) / j



4. (10%) Draw the threaded representation of the following binary tree.

-1



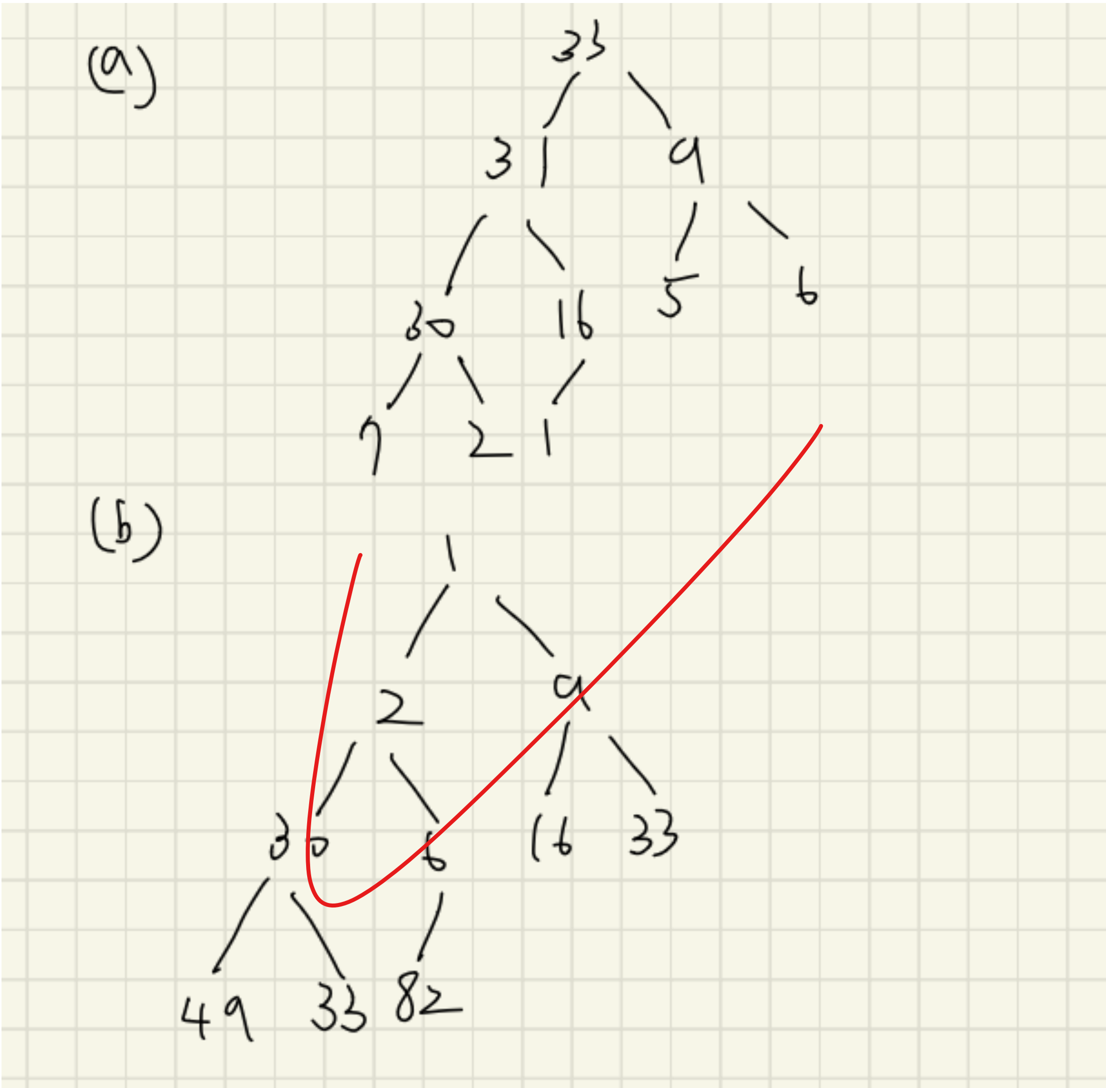


5. (10%) Suppose that we execute the following instructions:

push 7, **push** 9, **push** 16, **push** 30, **push** 49, **pop**, **push** 82, **push** 5, **push** 33, **push** 31, **push** 6, **pop**, **push** 2, **push** 1.

(a) What is the resultant heap if we use a max heap?

(b) What is the resultant heap if we use a min heap?

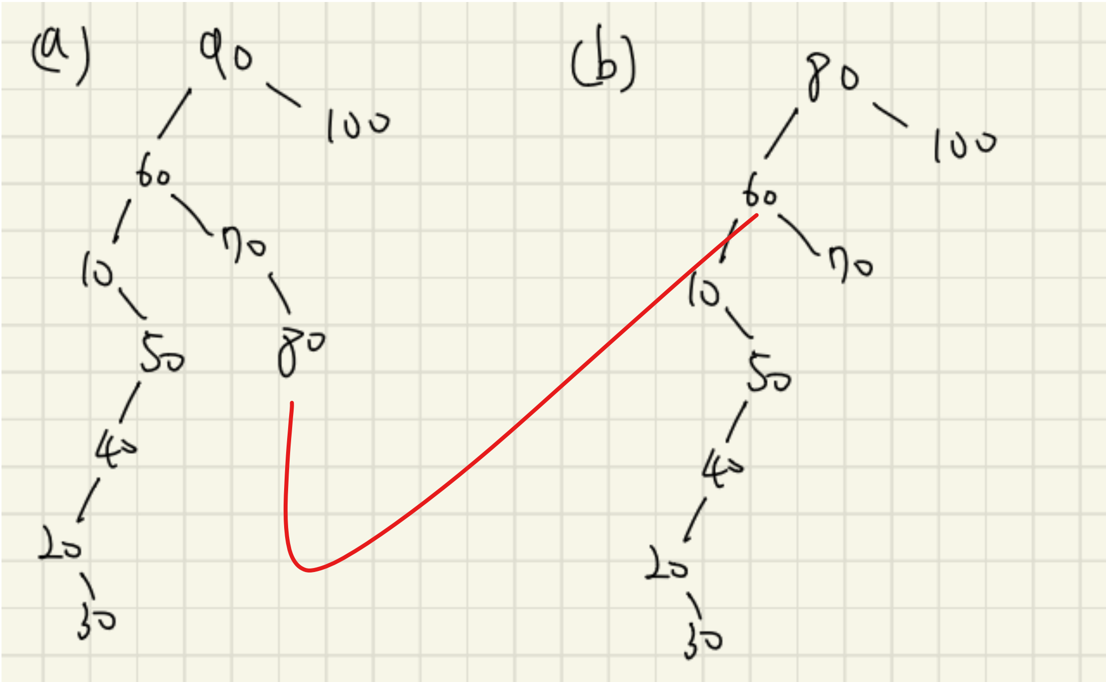


6. (10%) Given the following numbers:

90, 60, 70, 100, 10, 50, 40, 20, 30, 80.

(a) Construct a binary search tree.

(b) From (a), what does the binary search tree become if we delete 90 using the rule "replaced by the largest element in its left subtree"?

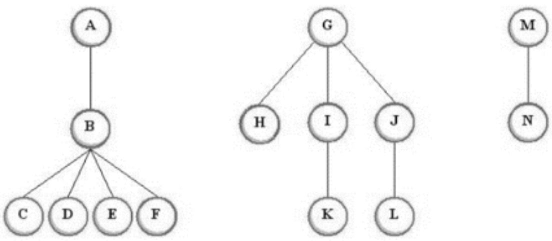


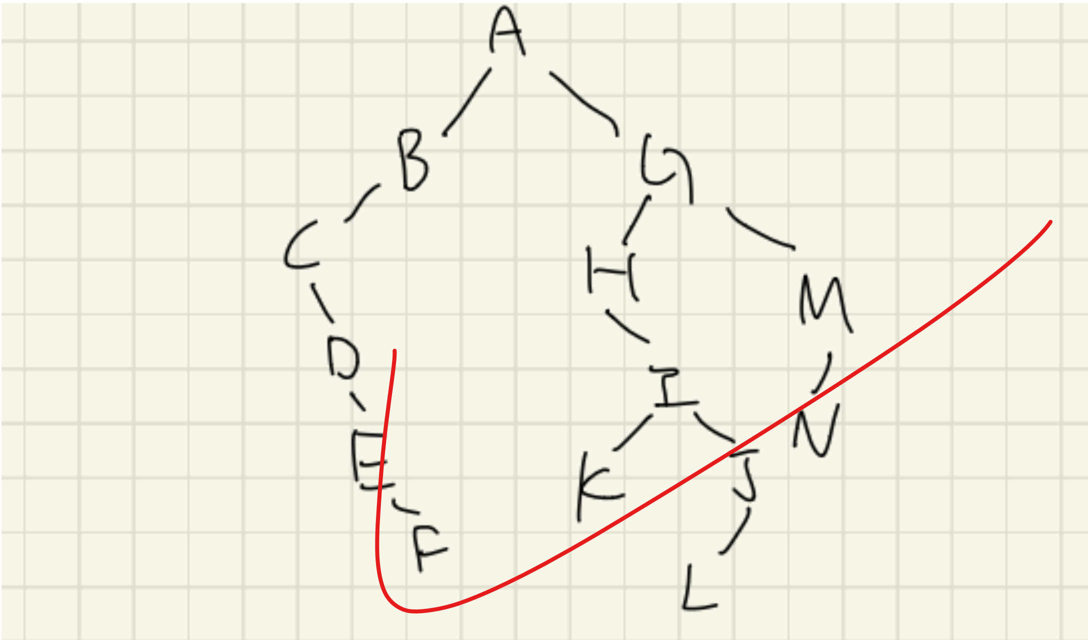
7. (12%) How many distinct 6-node binary tree structures can we create?

$$\frac{1}{n+1}\binom{2n}{n} = \left(\frac{12!}{6!\,7!}\right) = 132 \ \#$$

8. (13%) Transforming the forest into a binary tree using the rule:

- If $T_1, \ldots, T_n$ is a forest of trees, then the binary tree corresponding to this forest, denoted by $B(T_1, \ldots, T_n)$
- Root equal to root(T_1)
- Left subtree equal to $B(T_{11}, T_{12}, \ldots, T_{1m})$, where $T_{11}, \ldots, T_{1m}$ are the subtrees of root (T_1); and has right subtree $B(T_2, \ldots, T_n)$




A
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